

Program – 8: Create Kubernetes Pods using YAML Descriptors

Description

This program demonstrates deploying a Node.js application inside a Kubernetes Pod using declarative YAML instead of imperative commands.

Procedure

Step 1: Create Node.js App (server.js) const http =

```
require("http"); const port = 3000; const server =  
http.createServer((req, res) => { res.end("Hello from  
Node.js running inside Kubernetes!");  
});  
server.listen(port);
```

Step 2: Create ConfigMap (nodejs-

configmap.yaml) apiVersion: v1 kind: ConfigMap

metadata:

name: nodejs-app-config data:

```
server.js: | const http = require("http"); const port =  
3000; const server = http.createServer((req, res) => {  
res.end("Hello from Node.js running inside Kubernetes!");  
});  
server.listen(port);
```

Step 3: Create Pod (nodejs-pod.yaml)

```
GNU nano 8.3 nodejs-pod.yaml *
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs-app
spec:
  containers:
  - name: nodejs-container
    image: node:18
    workingDir: /usr/src/app
    command: ["node", "server.js"]
    ports:
    - containerPort: 3000
    volumeMounts:
    - name: nodejs-app-volume
      mountPath: /usr/src/app
  volumes:
  - name: nodejs-app-volume
    configMap:
      name: nodejs-app-config
```

Step 4: Apply Files `kubectl apply -f nodejs-`

`configmap.yaml kubectl apply -f nodejs-`

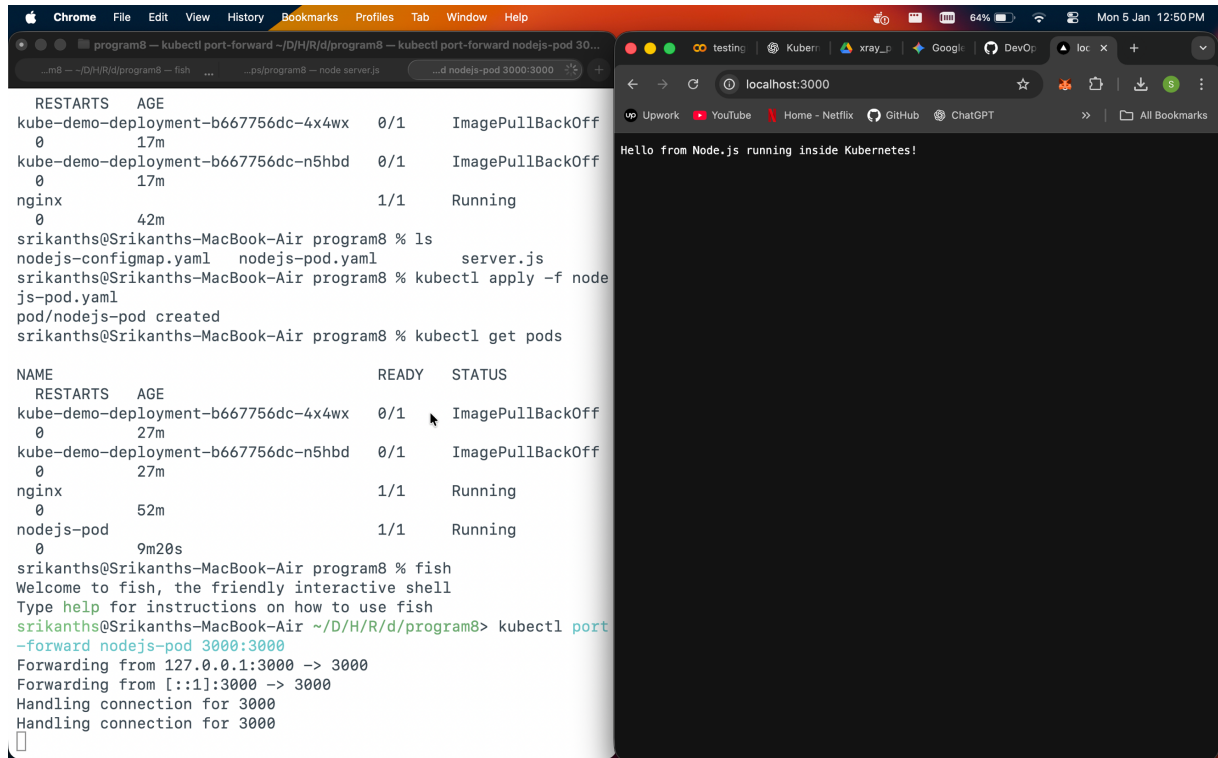
`pod.yaml` **Step 5: Port Forward** `kubectl port-`

`forward pod/nodejs-pod 3000:3000`

Open: <http://localhost:3000>

Expected Output

Hello from Node.js running inside Kubernetes!



The screenshot shows a terminal window on the left and a web browser on the right. The terminal window displays the following commands and output:

```
srikanths@Srikanths-MacBook-Air program8 % ls
nodejs-configmap.yaml  nodejs-pod.yaml  server.js
srikanths@Srikanths-MacBook-Air program8 % kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
srikanths@Srikanths-MacBook-Air program8 % kubectl get pods
```

NAME	RESTARTS	AGE	READY	STATUS
kube-demo-deployment-b667756dc-4x4wx	0	27m	0/1	ImagePullBackOff
kube-demo-deployment-b667756dc-n5hbd	0	27m	0/1	ImagePullBackOff
nginx	0	52m	1/1	Running
nodejs-pod	0	9m20s	1/1	Running

```
srikanths@Srikanths-MacBook-Air program8 % fish
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program8> kubectl port-forward nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```

The web browser on the right shows the address bar with "localhost:3000" and the page content "Hello from Node.js running inside Kubernetes!".

Result

Successfully deployed a Node.js application inside a Kubernetes Pod using YAML descriptors.